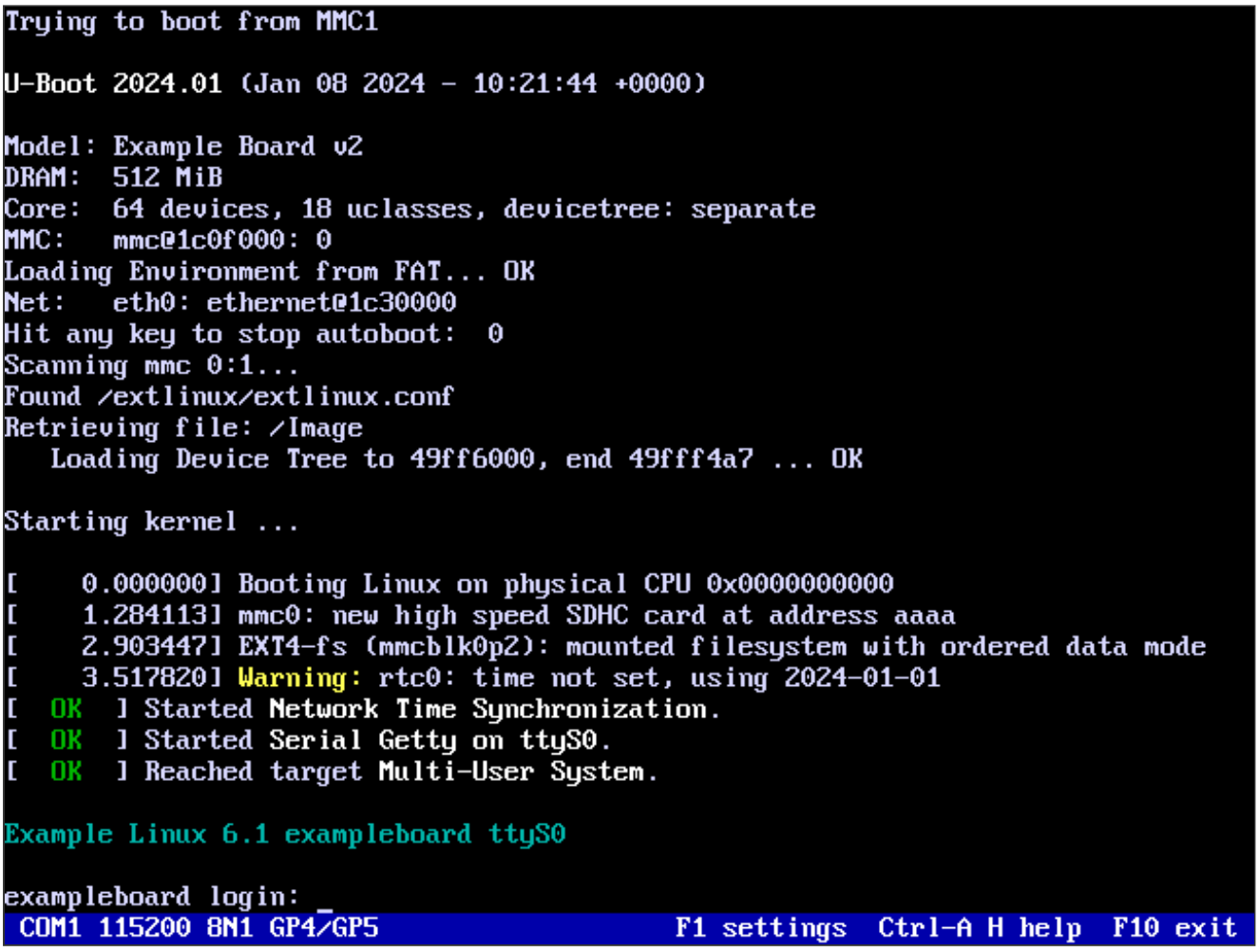
Pico 2 mini OS Serial Terminal

Pico 2 mini OS · October 27, 2026



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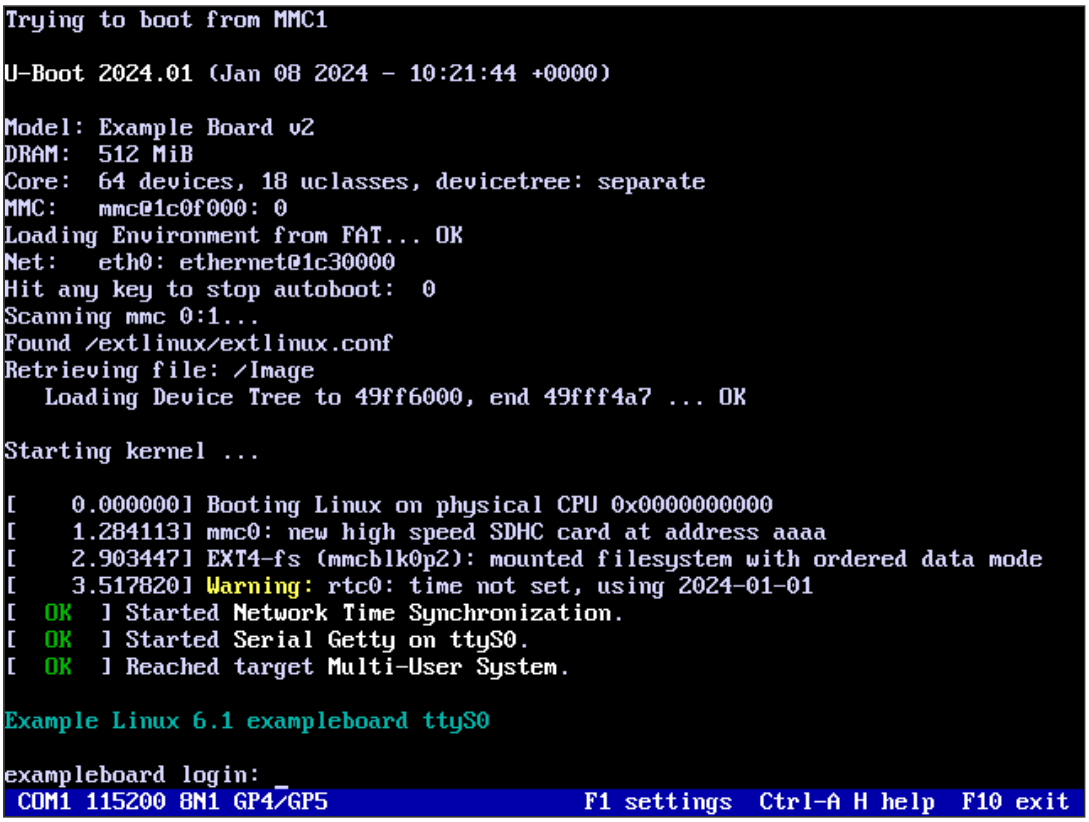
`term [baud] [format]` (or `serial`) is a terminal on the COM1 serial port, with a settings page, logging and file transfer by XMODEM, YMODEM and ZMODEM.

The terminal

`term` (or `serial`) is a terminal on COM1, the serial port BASIC also uses: UART1, TX on GP4 and RX on GP5 unless the settings say otherwise (3.3 V levels; connect the far end's RX to TX, TX to RX, and ground). The debug port on GP0/GP1 is never used.

```
term                 the saved settings (115200 8N1 at first)
term 9600            another speed this time
term 9600 7E1        speed and format
```

What arrives goes through the console's VT100/ANSI handling, so colours and cursor movement from the far end work. Rows 1-29 are the terminal (set the far end to 29 rows, e.g. `stty rows 29`); row 30 is a status line with the port settings and key hints. What arrives also goes to the debug serial port when it can keep up.



Above: a board's boot messages arriving on COM1 at 115200 baud, colours and all. (The sample session is made up; any device with a serial console looks much the same.)

Key	Action
F1, or Ctrl-A S	Settings page
F10, or Ctrl-A X	Leave the terminal
Ctrl-A B	Send a BREAK (250 ms)
Ctrl-A C	Clear the screen
Ctrl-A E	Local echo on / off
Ctrl-A L	Log what arrives to a file (default `TERM.LOG`, added to); again to stop

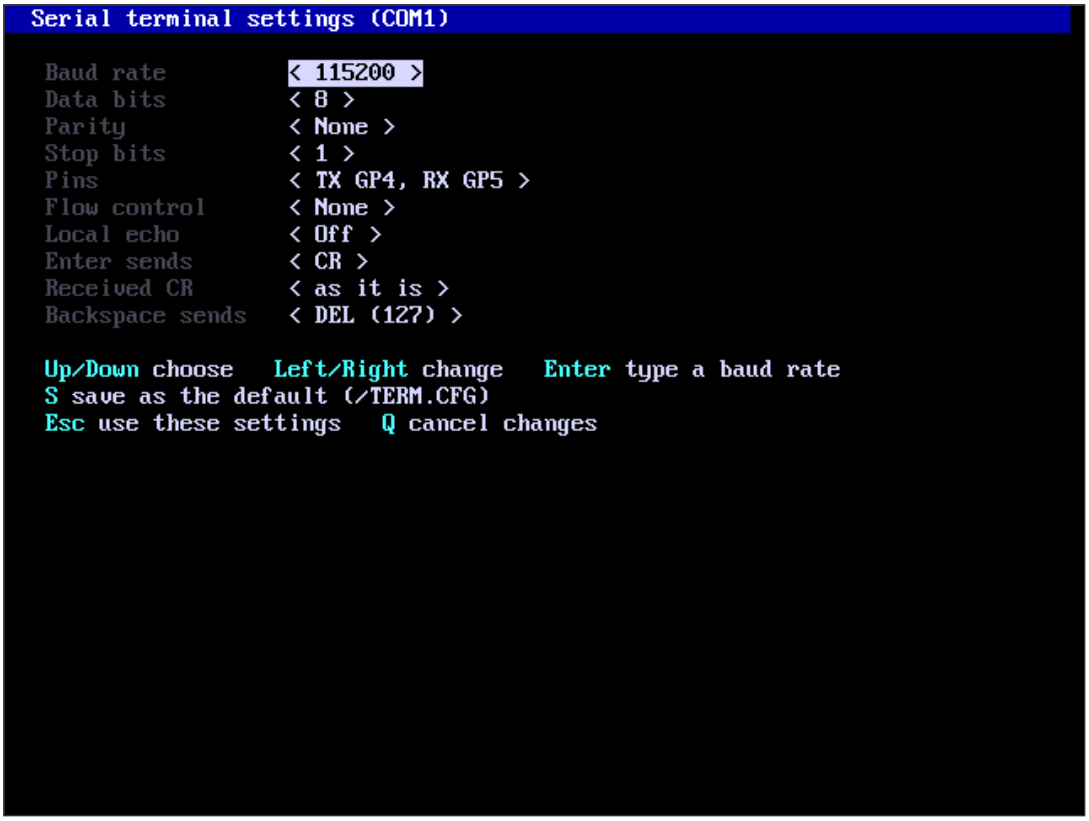
Key	Action
Ctrl-A F	Send a file as it is (Esc stops)
Ctrl-A U	Upload: send files with XMODEM, XMODEM-1K, YMODEM or ZMODEM
Ctrl-A D	Download: receive files with XMODEM, YMODEM or ZMODEM
Ctrl-A H	Key help page
Ctrl-A Ctrl-A	Send Ctrl-A

Other keys go to the far end; arrows, Home, End, Delete and function keys as VT100 sequences.

The settings page (Up/Down choose, Left/Right change):

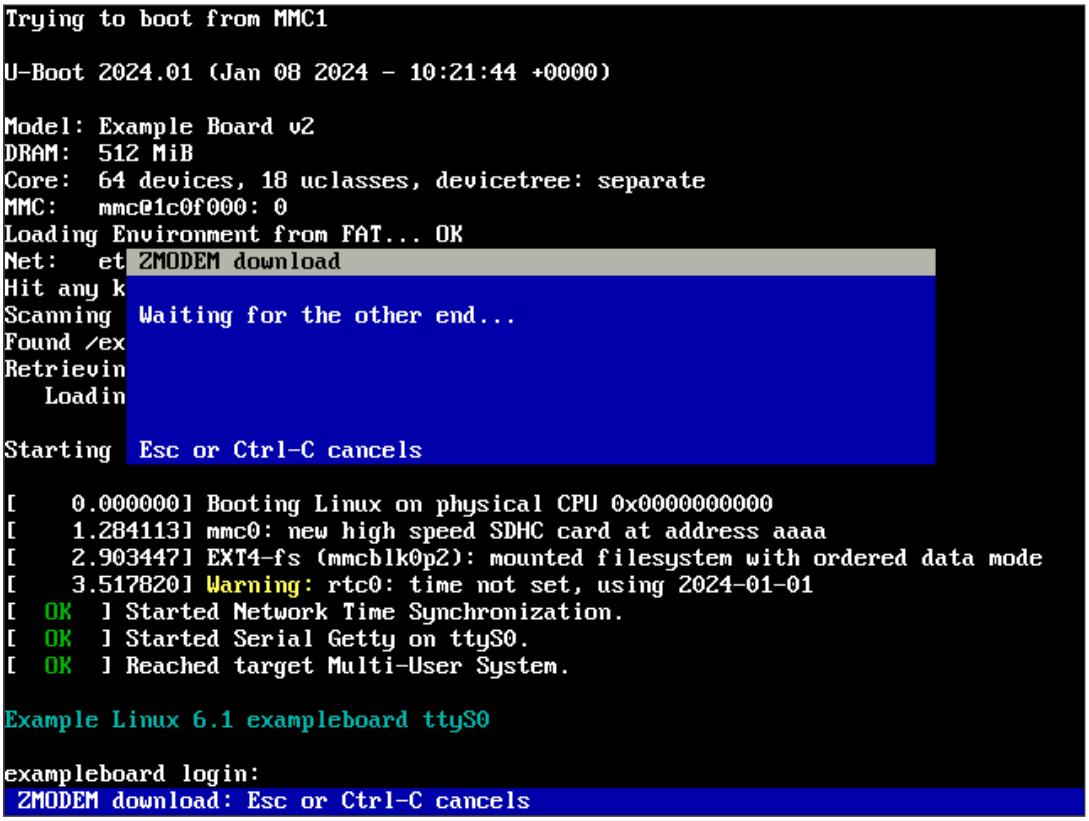
Setting	Choices
Baud rate	300 to 921600 from a list, or Enter to type any rate from 50 to 4000000
Data bits	5, 6, 7, 8
Parity	None, Even, Odd
Stop bits	1, 2
Pins	TX/RX on GP4/GP5, GP8/GP9, GP20/GP21 or GP24/GP25
Flow control	None, or RTS/CTS on the next two pins (CTS GP6 / RTS GP7 for GP4/GP5, and so on)
Local echo	Off, On
Enter sends	CR, LF, or CR LF
Received CR	as it is, or with a line feed added
Backspace sends	DEL (127) or BS (8)

`S` saves the settings as the default in `/TERM.CFG`, Esc uses them and goes back, and `Q` cancels the changes. Changing the port settings reopens the port.



File transfer: XMODEM, YMODEM, ZMODEM

Ctrl-A U sends files and Ctrl-A D receives them; a key picks the protocol. A window shows the file, bytes and percentage, speed, retries and files done; Esc or Ctrl-C cancels (the other end is told, with CAN characters).



Above: Ctrl-A D, then Z: the ZMODEM window waiting for the far end to start sending (`sz file` there). Once a file starts, the window shows its name, bytes, speed and retries.

Protocol	Sending	Receiving
XMODEM	one file, 128-byte blocks, CRC-16 or checksum (the receiver chooses)	one file; you type the name to save it as. Its size is rounded up to 128 bytes, padded with Ctrl-Z, as XMODEM has no sizes
XMODEM-1K (K)	one file, 1024-byte blocks	(the same as XMODEM)
YMODEM	several files with names and sizes	the sender's names; exact sizes
ZMODEM	several files, streaming	the sender's names; carries on with a partial file when the sender asks (`sz -r`)

- Files to send: type one or more names separated by spaces; `*` and `?` work in the last part, e.g. `EXAMPLES/*.BAS` (up to 32 files).
- Received files go in the current folder. A name that is taken gets `-1`, `-2` ... before the extension, so nothing is overwritten.
- A ZMODEM download starts by itself when the far end runs `sz file`: the terminal sees the start of ZMODEM and opens the window.
- ZMODEM uploads first send `rz` and Enter, which starts the receiver on a Unix shell.
- A broken XMODEM or YMODEM download is deleted; a broken ZMODEM one is kept so that `sz -r` can finish it.
- The receiver asks a ZMODEM sender to wait for an ACK every 4 KB, so a slow drive never loses data; RTS/CTS flow control also works.